Automatic vs Manual Investing: Role of Past Performance

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Abstract

Using unique data from a leading peer-to-peer (P2P) lending platform, we investigate the link between past investment performance and choice of auto-investing tool. Our results suggest that investors with poorly performing loan portfolios are more likely to switch automatically. This negative relationship can be explained by algorithmic aversion or investor inattention. In other words, the results suggest that good-performing investors who pay close attention to their loan portfolios or are not interested in using automated services are more likely to rely on themselves in manual mode. These results are robust to alternative specifications.

JEL Classification Codes: G11, G40, G51, D90

Keywords: FinTech; Peer-to-Peer Lending; Investor Switching; Automatic Bidding

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1. Introduction

In 2008, the first robo-advisor service was launched, which used algorithms to help investors with their investment strategies. Since then, the number of financial service providers offering robo-advisory services to their clients has increased dramatically, making passive rule-based investments more popular (Germann and Merkle, 2022).[1] The literature identifies several factors that could explain the demand for automated tools, including under-diversification (Loos et al., 2020), spending and saving behavior (D'Acunto and Rossi, 2021a), self-improvement and financial well-being (Rossi and Utkus, 2021), and financial literacy (D'Acunto and Rossi, 2021b). However, a more important factor could be the performance of the passive strategy, which could encourage the decision to leave the active investment strategy. This study aims to advance current knowledge by investigating how past portfolio performance affects the decision to switch to automation.

Portfolio allocations are a major concern for investors regarding decision-making. This is because a lack of diversification can lead investors to make suboptimal financial decisions. Thus, robo-advisors have been introduced to optimize portfolios. D'Acunto et al. (2019) show that under-diversified investors are more likely to benefit from robo-advising than their well-diversified counterparts. Similar results were obtained by Reher and Sun (2019), particularly for medium-sized investors. Moreover, individuals with limited self-directed investment knowledge, large cash holdings, and high trading volumes display significant gains from adopting robo-advising (Rossi and Utkus, 2020).

Peer-to-peer (P2P) lending markets are an ideal environment for our study. These lending platforms propose different frameworks for automated financial tools that help individuals make better financial decisions. Automated tools have gained rapid interest from lenders for several reasons: (i) time saving, (ii) good at decision-making, (iii) elimination of human biases, and (iv) ease of use. This set of additional features makes this market an appropriate environment for our investigation. First, P2P lending markets have lower transaction costs and margins charged by a platform than traditional banks. Second, investors with access to a wide range of lending opportunities have low information costs, each providing detailed information about the borrowers. Finally, entry and exit costs are low, which can attract borrowers and lenders. These features make the P2P lending market suitable for comparing self-directed and automatically managed investment decisions.

[1] See "The end of active investing?" (Financial Times, January 19, 2017) (Available at: https://www.ft.com/content/6b2d5490-d9bb-11e6-944b-e7eb37a6aa8e), accessed on Aug 30, 2022.

In this study, we used detailed data of biddings on listings from a leading P2P lending platform in China, Renrendai.com. The data span from 2010 to 2018 and have several unique aspects. First, the platform allows investors to access hybrid investment strategies. This helps identify investors who switch from one bidding mode to another. Second, human investors using automated tools cannot observe borrower characteristics, which allows us to identify whether any type of discrimination is being applied. The entire sample accounts for 890,000 users, with 94% of their investments attempted using the auto-bidding tool. Additionally, hybrid users, who accounted for 68,000 investors, attempted 82% of the automatic bids. Users of this platform are highly dependent on automatic lending, which confirms the popularity of this service.

Our initial findings suggest that individual underperformance in the past, measured using the weighted average Sharpe ratio, is associated with the use of an auto-bidding toolbox. We also found that this automated tool does not discriminate against borrowers based on gender, marital status, financial literacy, or employment characteristics. Moreover, our results show that lenders that performed poorly in the past are likely to switch to the automated mode by 1.4 percentage points. This relationship may be driven by algorithmic aversion and investor inattention. In the latter, good performers who pay attention to the market prefer to rely on themselves. In addition, building trust with the auto-bidding toolbox entails considerable effort, as users' interest in automation seems to fade rapidly. When testing for time pressure using the fast decision-making indicator, we found no relationship between time pressure and switching to automation. Finally, experience can play a role in investors switching to manual mode as they become more confident in attempting their bidding ability.

This study is related to several strands of the literature. First, it contributes to the literature on the adoption of automated financial tools. In particular, prior studies have addressed the importance of algorithmic choices in terms of investments (D’Acunto et al., 2019; Rossi and Utkus, 2020; Reher and Sun, 2019; Ge et al., 2021), consumption savings and debt management (D’Acunto and Rossi, 2021a), durable investing (Gargano and Rossi, 2020), and cultural biases (D’Acunto et al., 2022). Our study contributes to the literature by documenting the impact of past portfolio performance on the decision to switch to an auto-bidding tool and exploring other channels, such as investor inattention, time pressure, experience, and algorithmic aversion.

Our setting differs from those of existing studies, which are important for understanding investor behavior. In our study, investors using the auto-bidding toolbox could change between bidding modes (manual or automatic) without restrictions. By contrast, Rossi and

Utkus (2020) analyzed a modern robo-advising tool for indexed mutual funds, where once individuals sign up for the service, they cannot return to a human financial advisor unless they cancel the contract. D'Acunto et al. (2019) investigate a portfolio optimizer tool that only provides stock recommendations to investors, whereas the tool used in the experiment can invest on behalf of lenders. Furthermore, Reher and Sun (2019), Ge et al. (2021), and D'Acunto et al. (2022) employed cross-sectional data, whereas our study uses a large sample at the investor bid level.

Second, this study relates to the literature on algorithmic aversion. Several social psychology studies have shed light on the level of human decisions that lead to conflict *versus* automated decisions (Castelo et al., 2019; Logg et al., 2019). Dietvorst et al. (2018) presented an algorithmic model adopted by participants and discussed the possibility of reducing algorithmic aversion. The authors found that individuals who were able to modify their forecasting algorithms were more satisfied with the process than those who did not modify the forecasting. We extend the existing literature by providing evidence on how lenders trust automated services after having tried them multiple times.

Finally, we supplement the existing literature that investigates the active versus passive investment strategy and performance (Stoughton et al., 2011; Stambaugh, 2014; Easley et al., 2021; Coles et al., 2022; Gârleanu and Pedersen, 2022). We contribute to the literature by showing that investors' underperformance explains their interest in passive rule-based investing such as automated solutions or robo-advising.

The remainder of this paper is organized as follows: Section 2 introduces the Renrendai.com platform and the auto-bidding tool; Section 3 introduces our data and sample collection process; Section 4 reviews the methodology and introduces the econometric model used; Section 5 presents our main results; and Section 6 concludes the paper.

2. Renrendai.com platform and automation

Renrendai.com is a leading P2P lending platform (Caglayan et al., 2020a) that was founded in 2010 (Liao et al., 2021). It has an AAA rating, the highest rating that the Chinese Academy of Social Sciences can provide for P2P lending platforms. By 2015, the platform had more than two million registered users and was among China's top 100 Internet companies. In 2018, Renrendai.com had more than one million approved loans and more than 10 billion RMB in total investments. This platform no longer exists, as it shut down in 2021 owing to regulations issued by the Chinese government that made P2P lending platforms struggle.

For borrowers to apply for credit loans ranging from 3,000 RMB (ca. 450 US$) to 500,000 RMB (ca. 72,250 US$), they must provide information such as marital status, monthly income, educational level, gender, and other personal details. After submitting all these documents, the platform evaluates the borrower's application and assigns customers a credit rating as follows: AA, A, B, C, D, E, or HR (high risk). These ratings allow investors to determine a borrower's level of riskiness.

In contrast, it is easy for investors to get verified by the platform. Bidders only need to register with the platform to complete the verification process. After registration, they begin searching for suitable loans for funding.

Using this platform, bidders can choose an investment strategy. This strategy can be based entirely on self-directed bidding, automated bidding, or a hybrid method that combines both investment modes. For example, if investors decide to use the manual mode, they bid on loans using information that satisfies their preferences, such as bidding on a loan with the desired characteristics. Alternatively, if automated services are used, there are no options for analyzing information at the time of the bids. Instead, a goal is set, and bidding is carried out using loan parameters. The only factors that an investor can consider are the bid amount to be invested, maturity, and interest rate when investing in an auto-bidding toolbox.

Reliance on machines has become popular in the financial industry over the past decade.[2] Equity investors can rely on machines for active investments, where machine learning algorithms identify outperforming stocks.[3] A similar change has been occurring in P2P lending platforms, and automated solutions are becoming increasingly attractive to lenders.

Renrendai.com has two unique features related to auto biddings. First, the platform helps lenders carry out decentralized bidding and recurring lending according to the lender's bidding scope, with no access to borrower characteristics. Second, it allows for human interference if it is not satisfied by the machine.

The automated-bidding tool is popular among investors on Renrendai.com and consists of three services. The first, the preferred service, can provide investors with 12 months of continuous automatic bidding for a specific fee. The second service consists of a fixed-term contract that expires 3, 6, 9, or 12 months after signing the contract. The third service offers a fixed monthly investment during the first year. The choice of services depends only on

[2] Popular platforms, such as etoro in the UK and Robinhood in the US, provide automated services for their clients.

[3] See "Will bots replace humans in active equity investment?" (Financial Times, October 02, 2019) (Available at: https://www.ft.com/content/efe4f97a-adb1-3cea-b098-2b616b5ce531), accessed on Aug 30, 2022.

investors’ preferences, and the differences between the services are mainly with regard to the duration of the service and the bidding amount by investors.

3. Data Description

We collected data from Renrendai.com between October 2010 and October 2018. On this platform, 892,716 lenders have placed more than 75 million bids on approximately 0.6 million listings. First, we collected information on the loans and borrowers for each approved application. Second, we gathered investor-level data based on each bid's timestamp, the amount invested in each loan, and the bidding method used to fund each loan. By combining investor- and listing-level data, we obtained a sample of approximately 75 million observations at the investor-bid level. Each loan application has financial information such as maturity, interest rate, and loan riskiness, as well as borrower characteristics such as employment, marital status, educational level, and gender. Our study focuses on investors who conduct both manual and automated bidding. The final sample included 892,716 investors, with 824,686 deciding to use only one bidding method, either automatic or manual, and 68,020 investors deciding to switch from one bidding method to another.

[Table 1 Here]

Panel A of Table 1 provides the basic summary statistics for the key variables in this study. The average number of automatic bids is 89%, with a standard deviation of 31%, implying that some users rely heavily on the auto toolbox, whereas others try to limit their usage to this service. *Share Auto Bidding* represents the share of automatic bidding held by investors on the platform, and we find that 82% of automatic bids are recorded in lenders' portfolios, on average. The high share of auto bidding confirms the increasing popularity of robo-investing in P2P lending platforms.

Popular P2P platforms in China that have introduced robo-investing tools, such as PPdai.com, have shown that investors are heavily dependent on automation. Ge et al. (2021) documented that the number of bids attempted by machines outnumbered those attempted by lenders on PPdai. The average age of the investor profiles that show the activity of the investors on the platform can last up to 106 days (in logs, 4.67), and the mean volume of the bids is approximately 292 bids attempted (in logs, 5.68) by users. Panel B of Table 1 shows the investor-level statistics. There are no major differences from those reported in Panel A. However, it is worth mentioning that 4% of lenders switch to automatic and 2% switch to manual.

Panel C of Table 1 reports the summary information for approximately 842,000 loan observations. In the sample, the average maturity is just over 30 months, implying that these loans have a long-term maturity. Additionally, the average interest rate earned on loans is 11% and the average amount requested is 59,000 RMB (approximately 8,535 US$).

Panel D of Table 1 reports the summary statistics for the sample at the borrower level. The data show that, on average, 58% of the borrowers are married, 1% are highly educated, 66% are male, and 18% are self-employed. In addition, borrowers have a risk score of 69 (in logs, 4.27), meaning that on average, they are not high-risk borrowers.

4. Methodology

4.1 Sharpe ratio calculation

We constructed a measure for portfolio performance using investors' Sharpe ratios, which change at the time of each bid depending on historical performance (Guo et al., 2016). We modified the Sharpe ratio, where the return is the interest rate and the risk is the credit score. The credit score on Renrendai.com is a numerical value ranging from 0 to 245, where 0 is the riskiest and 245 is the safest investment. We modified the credit score, such that the higher the score, the riskier the loan. We created a maximum credit score with a value of 246 and then subtracted the maximum credit score from the original credit score provided by the platform.[4]

$$\text{Risk}_j = \text{Max Credit Score}_j - \text{Credit Score}_j \tag{1}$$

where risk measurement is the difference between the maximum credit score on the platform and the credit score for each loan *j*. The risk measurement was set such that the higher the credit score, the riskier the investment. As a result, the risk score shows that the more it deviates from 1, the riskier the investment. Additionally, we weighed both return and risk with investor bid amounts to obtain a plausible measure of portfolio performance. We computed the average Sharpe ratio across successful bids within a lender, weighted by the bid amount.

4.2 Econometric Specification

Our aim is to investigate the relationship between past portfolio performance and switching to an automatic-bidding tool. We begin our analysis by establishing the characteristics of the

[4] The highest score is 245, as mentioned previously; hence, this score is only greater by one unit.

loans and borrowers who choose the auto-bidding tool by estimating the following regression: [5]

$$\begin{aligned}\mathrm{AutoBidding}_{i,t} = {} & \beta_0 + \beta_1 \mathrm{SharpeRatio}_{i,t-1} + \beta_2 \mathrm{ShareAutoBidding}_{i,t-1} \\ & + \beta_3 \mathrm{ShareAutoBidding}^2_{i,t-1} + \beta_4 \mathrm{Loan}_{i,\gamma} + \beta_5 \mathrm{Borrower}_{i,\gamma} + \varepsilon_{i,t}\end{aligned} \tag{2}$$

where *AutoBidding* is a dummy variable that takes the value of 1 if the bid *t* of investor *i* is attempted by automated services and 0 if it is attempted manually. $SharpeRatio_{i,t-1}$. is the portfolio performance measurement for investor *i*, lagged by one bid attempt. $ShareAutoBidding_{i,t-1}$ is the share of automatic bidding of investor *i* at an on-bid attempt lag. $Loan_i$ is a vector of loan characteristics that includes the logarithm of one plus maturity, interest rate, and the logarithm of one plus loan amount. $Borrower_{i,\gamma}$ is a vector of borrower characteristics that includes dummy variables for self-employment, marriage, higher education, gender, and borrower risk score. By including borrower characteristics, we argue that automation is likely to reduce discriminatory practices against a specific group of individuals. δ_i and D_i are the investor and hour of bid fixed effect, which we use instead of the loan and borrower characteristics. The standard errors are robust and clustered at the individual investor within a loan level.

As a next step, we attempt to establish the factors that determine the change in bidding mode, from manual to automatic or vice versa, and estimate the following regression:

$$\begin{aligned}\mathrm{Switch}\{G, A, M\}_{i,t} = {} & \beta_0 + \beta_1 \mathrm{SharpeRatio}_{i,t-1} + \beta_2 \mathrm{ShareAutoBidding}_{i,t-1} \\ & + \beta_3 \mathrm{ShareAutoBidding}^2_{i,t-1} + \beta_4 \mathrm{SharpeRatio}_{i,t-1} * \mathrm{ShareAutoBidding}_{i,t-1} \\ & + \beta_4 \mathrm{SharpeRatio}_{i,t-1} * \mathrm{ShareAutoBidding}^2_{i,t-1} + \delta_i + D_i + \varepsilon_{i,t}\end{aligned} \tag{3}$$

where $Switch\{G, A, M\}_{i,t}$ is a dummy variable that takes the value of one if investor *i* switches its mode at the bid attempt. However, we distinguish between the three investor choices and employ three dummy variables. The first binary variable indicates *General Switch*$_{i,t}$ which equals 1 if a lender switches to either manual or automatic mode, and 0 otherwise. The second variable is *Switching to Automatic*$_{i,t}$ and equals 1 if investors switch to automatic, 0 otherwise. The third variable is *Switching to Manual*$_{i,t}$, which is a dummy variable that equals 1 if investors switch to the self-directed mode and 0 otherwise. As above, δ_i and D_i are the investor and hour of bid fixed effect, which we use instead of the loan and borrower characteristics. The standard errors are robust and clustered at the individual investor within a loan level.

[5] We choose a linear probability model instead of a logit model because of the sample size and because we do not seek to predict probabilities.

5. Results

5.1 Descriptive statistics

We start our analysis by splitting the sample into two categories that only control for manual and automatic bids and show the t-test mean difference between the two bidding modes. Table 2 reports the means and standard deviations of the two subsamples. We observe that a higher Sharpe ratio is achieved using automated bids (0.16) than using self-directed bids (0.15). Users with machine-made bids are more likely to be active and bid more on the market than those who rely on manual bids. When inspecting the inattention indicator *Log (Decision Time)*, the auto-funding tool seems to be more aware of the information flow in the market because once a loan is issued, it instantly attempts a bid to fund the loan. Moreover, on average, manual users prefer to see the loan halfway funded before starting bidding (0.52%). Users who rely on automation start funding the loan (on average, 0.49%) before being half-way funded. Automation is also associated with long-term investments, as indicated by *Maturity*. Machines are known to prefer to invest in long-term projects. Additionally, the auto-investing tool on renrendai.com places bids on lower-risk borrowers, with a mean of 4.21. Consequently, automatic users can invest in loans that are less profitable than those for manual users. In addition, automatic users have a higher probability of investing in unmarried female borrowers.

[Table 2 Here]

5.1 Usage of automated biddings

Table 3 presents the results of determining the usage of the automation option. We find that the Sharpe ratio coefficient is negative and statistically significant, at least at the 5% level, for all specifications. Thus, the results indicate that investors are more likely to invest in machine-made bids than in self-directed bids if they have underperformed previously. These results are also economically significant. In Column 4, the results show that a one-unit increase in the risk-adjusted interest rate for the Sharpe ratio increases the use of automated biddings by 0.2 percentage points. This result is not surprising, as Rossi and Utkus (2020) and Hao et al. (2022) show that robo-advising investments outperform self-directed accounts. They measured investment performance using portfolio returns, volatility, and the Sharpe ratio, and reported that robo-advising improves investment performance mainly by lowering portfolio volatility. Thus, the results are in line with those of D'Acunto et al. (2019), who found that investors better diversify their stock portfolios after adopting robo-advising. Moreover, Ge et al. (2021) found that investors adopting an automated financial tool in a P2P lending platform were inspired by encountering more loan defaults in the past. Overall, the results show a positive effect of robo-advising on investment performance compared to self-directed accounts, which explains why investors choose robo-advising.

Our results show that loan characteristics play a role in investors' choices of automated bidding. In all specifications, the coefficients of loan maturity are positive and statistically significant at the 1% level. This implies that individuals who rely on automation invest in higher-maturity loans. This can be explained by the fact that automated services prefer long-term passive projects (Menkveld, 2013).[6] The results also suggest that the auto toolbox is more likely to be associated with more profitable loans, as the coefficient for the interest rate is positive and statistically significant in Column 1. However, in Columns 1 and 2, the coefficients change sign when we introduce borrower and investor fixed effects into the regression. In other words, machine-made bids are considered less profitable investments when controlling for investors' and borrowers' heterogeneity. In our opinion, this effect occurs because investors and borrowers can control the interest rates. Borrowers are the ones who set interest rates, and investors, before using the auto-bidding toolbox, can set an interest rate target on the system. Thus, the automated financial tool on this platform bids for less risky and more profitable loans. However, when investors interact with this automated tool, they appear to set lower interest rates.

In Column 1, the results show that women, the unmarried, and self-employed individuals are likely to be funded by the auto toolbox. The coefficients for married and male workers are negative, whereas those for the self-employed are positive. All three coefficients were statistically significant at the 1% level. In contrast, we find that the coefficient for higher education is statistically insignificant.

Existing literature shows that human investors prefer to fund financially literate (Caglayan et al., 2020a), married, and male individuals (Chen et al., 2020). Additionally, human investors would usually show high uncertainty when evaluating an application made by a female applicant (Duan et al., 2020). Overall, this is related to the fact that human investors are more prone to bias (see, e.g., Foerster et al., 2017). Our findings partly confirm the existing results but also show that automatic bidding is likely to fund borrowers with whom human investors discriminate.

These results are consistent with the findings of D'Acunto et al. (2022), who provide evidence of how investors in micro-lending markets face significant losses when discriminating against a particular group of borrowers. Their results show that investors who

[6] In their paper, they report that the majority of HFT trades are considered passive trades (by 78.1%)

self-direct their investments tend to make investment mistakes, such as lending to borrowers who share the same religion and those with a higher social class. When investors adopted the automated-bidding tool, these behavioral biases started to be corrected, resulting in lower default rates (by 32%) and higher returns (by 11%). Finally, machine-made bids prefer borrowers with lower risk scores, as evidenced by the negative significance of the borrowers' risk coefficient. This might not be preferable for some individuals because human investors are not risk averse and would like to be involved in riskier investments to maximize their profits.

[Table 3 Here]

5.2 Decision to Switch

Next, we investigate the relationship between past portfolio performance and switching behaviors. Investors can switch between bidding modes by switching to automatic or manual modes when lending. As we are interested in the switching behavior of investors, we exclude users who rely on one bidding mode, either manual or automatic.

Column 1 of Table 4 presents the results of the decision to switch from one bidding mode to another. We find that the coefficient of the Sharpe ratio is positive and statistically significant at the 5% level. This means that investors are likely to make general switches when they have better risk-adjusted portfolio performance. To further investigate the switching behavior, we split switching into two indicators: switching to automatic and switching to manual. This allows us to investigate how past portfolio performance affects users' decisions to switch to an automated investment tool.

Column 2 presents the results of the decision to switch from manual to automatic. In contrast to previous results, we find that the coefficient of the Sharpe ratio is negative and statistically significant at the 1% level. In other words, investors with poor portfolio performance are more likely to switch automatically. The results indicate that a one-unit decrease in the weighted average risk-adjusted interest rate increases the chance of switching to an automated toolbox by 1.4 percentage points. This implies that lenders prefer changing their behavior to adopting automation due to a bad spell of poor performance.

In a related study, Rossi and Utkus (2020) find that investors with lower returns and higher risk sign up for robo-advice. Capponi et al. (2022), D'Hondt et al. (2020), and Reher and Sun (2019) show that automated financial tools can improve individuals' portfolio performance. Loos et al. (2020) show that investors who use robo-advising are likely to mitigate diversification by 11.7 percentage points when controlling for both time- and investor-fixed effects. Additionally, D'Acunto et al. (2019) found that adopting robo-advice benefits some investors because it increases their portfolio diversification and reduces portfolio volatility. Simultaneously, robo-advisors do not improve the performance or volatility of the portfolios of already diversified investors. On average, they find that investors have a better-diversified portfolio by 0.16 units, which is approximately 1.3% of the median number of stock investors held before using the portfolio optimizer. Therefore, underperformers believe that switching to the automated mode is better for them, as it might result in better financial outcomes.

[Table 4 Here]

In contrast, good-performing investors prefer to rely on themselves in the manual mode. In Column 3, we find a positive coefficient of the Sharpe ratio that is statistically significant at the 1% level. The coefficients indicate that a one-unit increase in the weighted average Sharpe ratio leads to a 1.8 percentage point increase in switching to the manual mode. This implies that users who perform well in the automated mode are likely to drop this service and rely on themselves in the manual mode. One might argue that individuals want to get involved in riskier loans, and hence achieve higher returns when funding loans. As reported in Tables 1 and 2, the automated toolbox bids for less-risky investments.

We investigate the sensitivity of our results using an alternative performance measure: the number of unique successful investments. Table A3 shows results that are consistent with those presented in Table 4. Therefore, the results confirm that underperforming investors are likely to switch automatically and those with better portfolio performance tend to switch manually.[7]

To further support this evidence, we run a cross-sectional regression at the loan level to explore the relationship between the loan shares of automatic bidding and loan performance indicators. The results in Table A5 suggest that if a loan is funded by a higher percentage of automated bidding, it is more likely to be associated with lower loan risks and interest rates.[8]

Thus far, the coefficients for auto-share bidding and the squared term are statistically significant for all specifications. Consequently, the results show that there is a nonlinear relationship between the coefficient of auto-share bidding and the three switching indicators. Figure 1 shows the predictive margins and illustrates the relationship between shares and auto bidding.

[7] These results are robust to several tests as we restrict the bid attempts to not less than 10 or 20 bids and still obtain consistent results, as presented in the Online Appendix: Tables A1 and A2. Further, we run the regression at the loan level (Table A4) and find consistent results to those reported in Table 3.

[8] These results are reported in the Online Appendix: Table A5. Further, similar results are reported at the investor-bid level in Table 3.

Panel A shows the relationship between the automated share usage and general switching. The figure shows an inverted U-shaped relationship. The results suggest that an increase in the number of automated bids can influence investors to change their behavior to a certain point (48%), after which the effect begins to decline. We also note that the Sharpe ratio distribution is similar to the auto-bidding share distribution, which might explain why investors decrease their bidding change. After holding 48% of automated shares, investors are less likely to consider switching bidding strategies because of their underperformance.

Panel B shows the relationship between share auto-bidding and automatic switching. The graph confirms a U-shaped relationship between the two variables. This illustration indicates that an increase in automated shares in investor portfolios reduces the possibility of relying on machines until automated shares reach 78%. [9]Furthermore, an increase in the shares of automatic bidding increases the probability of automatically changing the mode of investing. This means that the effect of share auto bidding fades after a while, and investors start to reconsider their options based on the automated shares they have.

An explanation for this result is that investors show interest in switching to automation as they receive performance benefits from relying on the automated tool at the beginning. However, after using the auto-bidding tool, their interest in switching to automation begins to decline because they assume that they can perform better on their own. Therefore, trusting the process between a lender and machine requires considerable time and effort.

Dietvorst et al. (2018) and Goll and Uhl (2018) provide evidence of an aversion to delegating tasks to machines. Castelo et al. (2019) demonstrated that humans reduce their algorithmic aversion when they rely on algorithms. This may occur because machines outperform humans and attempt to make better decisions (Kleinberg et al., 2018). At the same time, algorithmic advice allows humans to make wiser decisions (Logg et al., 2019), explaining why services based on algorithms are better for a specific type of user. However, the results of the present study indicate that this is not always the case.

Panel C presents the relationship between auto bidding for shares and switching to manual bidding. This figure illustrates an inverted U-shaped relationship, which contrasts with the results in Panel B. The observed relationship indicates that investors rely more on themselves as they hold more shares of automatic bidding until this effect starts to change when they hold 60% of automatic shares in their portfolios. Subsequently, individuals start to reconsider

[9] It is important to note that due to the large scale of the Sharpe ratio, the non-linear nature of the relationship may not be immediately apparent upon visual inspection of the figure. after 78%, there is a slight curve in share auto bidding, indicating a deviation from a linear relationship.

their choices by decreasing the probability of relying on themselves in the self-directed (or manual) mode. This finding might be due to algorithmic aversion, because when bidders feel uncomfortable with their reliance on automation, they prefer to maintain some degree of control over their investment decisions.

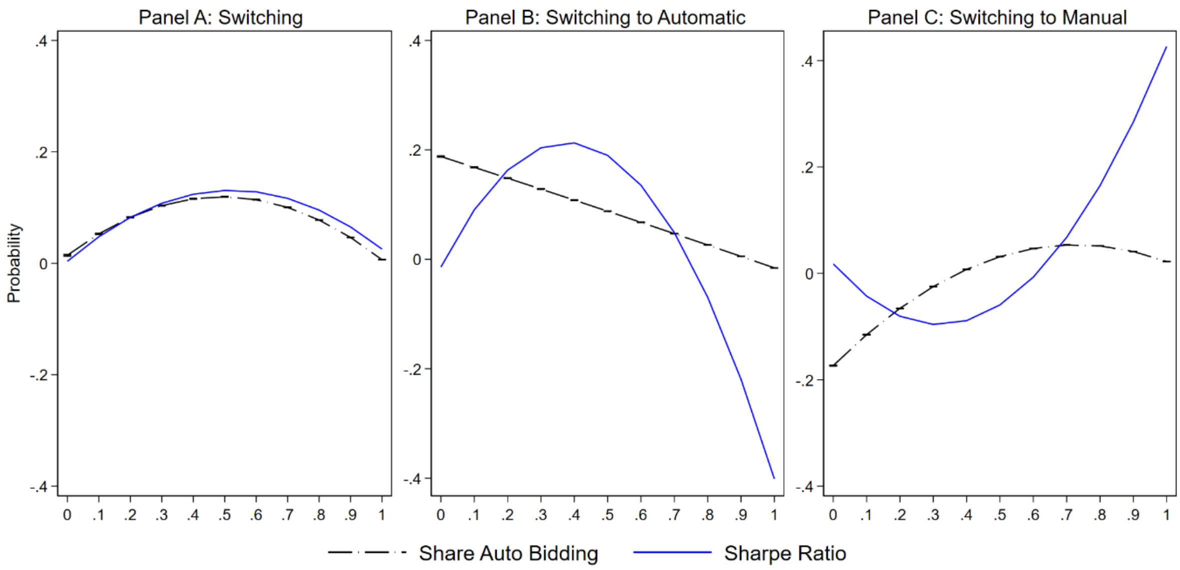

Figure 1: Switching Predictive Margins
This figure shows the predictive margins of switching indicators. Panel A shows the nonlinear relationship between share auto bidding and general switching. Panel B depicts the slight nonlinear relationship between share auto bidding and switching to automatic bidding. Panel C illustrates the nonlinear relationship between share auto bidding and switching to manual. The dashed-dotted black line in the three panels shows the relationship between share auto bidding and switching indicators. The long solid blue line represents the Sharpe ratio coefficients.

5.3 Investor inattention, time pressure, and experience

Thus far, our results show that investors are likely to switch bidding modes depending on their historical risk-adjusted portfolio performance. However, other factors may also play an important role in lenders' decision to consider automation. Thus, we verify the robustness of our results and investigate other factors that may determine investor decisions.

First, we analyze investor inattention during bidding. Caglayan et al. (2020b) show that lenders in P2P markets are simultaneously exposed to thousands of publicly available listings, which can lead to distraction. This can result in reliance on automated financial tools that do not face difficulties in processing millions of pieces of information. On Renrendai.com, the average loan completion time is less than 5 h (Caglayan et al., 2021). This implies that users who invest in loans in the first few hours are likely to pay close attention to loans listed on the market. Thus, we use the number of hours spent investing since listing a loan as a proxy for investors' inattention[10] and the flow of information during that hour using the hour of bid fixed effects.

Second, we explore whether time pressure affects changing modes of investment. Owing to physiological stress, time pressure can affect individual portfolio strategies. However, Kocher et al. (2013) presented a contradictory result by showing that there is no relationship between time pressure and preferences (e.g., risk attitudes toward gains). In our opinion, time pressure does not affect investors' switching investment modes. We follow Zhang et al. (2021) and proxy for time pressure using relative decision time.

The results in Table 5 show that the indicator of inattention, $Decision\,Time_{t-1}$, is positively associated with automatic switching. Overall, the coefficient proxying for inattention was statistically significant, at least at the 1% level, in all specifications. Moreover, the results are economically important, as we find that the number of hours spent attempting a bid by investors increases by 100% and switching to automation increases by 6.9 percentage points, ceteris paribus. We find that the interaction term between the Sharpe ratio and the proxy for inattention is positive and statistically significant at the 1% level. Thus, the results suggest that, as good-performing investors are more aware of the flow of information (e.g., listing loans) in the market, they are more likely to rely on themselves. In contrast, in Column (2), if the hours spent by individuals to make a decision decrease by 100%, switching to manual will increase by 4.5 percentage points, and the interaction term between the Sharpe ratio and the proxy for inattention is now negative and statistically significant at the 1% level.

D'Acunto et al. (2019a) find that investors pay close attention to their portfolios, regardless of the number of stocks held, even before adopting an automated financial tool. On a fast-moving platform, such as Renrendai.com, where investors are exposed to many loans with detailed information, it is easy to get distracted by the flow of information on the market and end up with poor-performing portfolios. This is where automation can play a role, given that the auto-investing tool can simultaneously robotize millions of portfolios. Investors who are more likely to become distracted can switch to automation as a possible solution. However, some individuals are less prone to being distracted and make good decisions, which is why they prefer to rely on themselves in the manual mode.

In investigating the effects of time pressure in Columns (3) and (4), we observe that the coefficient of the relative decision time is statistically insignificant. These results imply that time does not directly affect investors' bidding preferences. This result shows that a loan

[10] In our case, more hours spent to attempt a financial decision means that investors are more likely to be distracted and not paying attention to what is being listed on the market.

approaching the completion rate (being fully funded) and an investor changing their bidding behavior are not related. Investors can decide on their mode of investment at any stage of loan funding. This may be because investors in this market are less concerned about time pressure because not all individuals are informed or experienced. Some of these investors may be unprofessional lenders or have low bidding requirements. Therefore, time pressure has less of an impact on individual portfolio strategies, and psychological stress has a lesser influence.

[Table 5 Here]

Next, we believe that lenders' experience can directly impact the decision to rely on (or not rely on) machines. Investors who are active and spend more time in the market (more experienced) are more likely to be overconfident in the way they place their investments. We believe that this creates an environment in which overconfident investors are less interested in relying on algorithms. Chen et al. (2007) show that Chinese individuals seem to be overconfident investors, which explains their poor stock trading decisions. We follow the literature and use two proxies to measure investor overconfidence. In the first two columns, we use the age of the investor, and in the last two columns, we use the bidding volume.

The results in Table 6 indicate a positive relationship between the two proxies for overconfidence and manual switching. In Columns 2 and 4, the coefficients on overconfidence are positive and statistically significant at the 1% level. In Column 4, the results indicate that the more individuals bid, the more they switch to relying on themselves in the self-directed mode. These results align with the theoretical model of Odean (1998), who shows that the more investors trade, the more likely they are to be overconfident. In a later model, Gervais and Odean (2001) show that investors take too much credit for success. Therefore, investors overestimate success when they learn about their own abilities. This can cause investors to believe more in their decisions because they are overconfident, resulting in them relying more on themselves. Our results confirm their findings, and in economic terms, the results in Column 4 show that a 100% change in the number of bids attempted by lenders increases switching to manual by 0.5 percentage points, ceteris paribus.

At the same time, the interaction between overconfidence and portfolio performance does not seem to occur if investors consider switching to manual. In addition, our results show that the effect of overconfidence is not significant in automatic switching. Columns (1) and (3) show that the coefficients of overconfidence are statistically insignificant. This is because overconfidence is considered a human behavior and not a bias related to machines (Caglayan et al., 2021).

[Table 6 Here]

Finally, we find that the main variables in all specifications in Tables 5 and 6 have a similar effect as those presented in Table 4. Therefore, we infer that our main results are robust and unaffected by additional factors that may determine investment decisions. Hence, we can confirm that users who underperform based on their historical performance prefer to rely on machines, whereas others who have performed well in the past prefer to switch to the self-directed mode.

6. Conclusion

Recent technological developments have changed lending and borrowing from a process that goes through banks to one that can be easily accessed online. P2P lending platforms connect investors to borrowers by acting as financial intermediaries. These platforms are evolving by offering new services to customers that can ease their lending processes. As investors are changing their way of investing by adopting new technological developments that can save time and ease their investment processes, they are less interested in relying on traditional strategies. Recently, new automated solutions have emerged as suitable alternatives for human financial advisors. In fact, our study confirms the growing interest in automated solutions by investors, based on our analysis of the relationship between past loan portfolio performance and switching to automated biddings.

To conduct the investigation, we used data from Renrendai.com, a leading online P2P lending platform in China. Our results show that past underperforming investors were more likely to use automatic bidding than self-directed bidding. We find that the automated toolbox discriminates less against unmarried, female, and financially less literate borrowers. When focusing on hybrid users, we find that underperforming lenders are more likely to switch to automatic bidding, while good performers prefer to rely on themselves in the manual mode. This can be driven by other channels such as investor inattention and algorithmic aversion. We show that attentive individual investors with good portfolio performance prefer to rely on themselves in the manual mode. Thus, investors are temporarily reluctant to switch to automated tools. In addition, the estimates suggest that an interaction between investor overconfidence and reliance on machines is unlikely. However, overconfidence can play a role in lenders' investments.

Our results provide new insights into the P2P lending market and explain why an increasing number of individual clients use automated tools to make investment decisions. Therefore, the results are also important from a policy perspective, as the better performance of automated tools could provide greater stability to the financial system in the future. However, with the employment of automated tools on a greater scale, the related advantage(s) could disappear over time; whether this may be the case, we leave to future studies to explore.

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Table 1: Descriptive statistics for whole sample.

This table shows the mean (1), standard deviation (2), and quartiles (3)-(5) of the variables: Auto Bidding is a dummy variable that equals 1 if the investor uses automatic bidding, and 0 for manual biddings. Share Auto Bidding is the cumulative percentage of using automated bidding by investors. Share Auto Bidding^2 is the quadratic form of the share of automatic bidding. General Switch is a dummy variable that is 1 when investor switch to either automatic or manual, and 0 otherwise. Switch to Auto is a binary variable that indicates that 1 if automatic switch, and 0 otherwise. Switch to Manual is a binary variable where 1 if manual switch, and 0 otherwise. Sharpe Ratio is the risk-adjusted returns weighted by the bid amount. Profile Age is the logarithm of one plus the number of active days spent by an investor on the market. Bid Volume is the logarithm of one plus the number of bids attempted by investor. Decision Time is the logarithm of one plus the time (in hours) for an investor to place a bid on a loan. Relative Decision Time is the average of investor i's relative decision time for all bids. Maturity is the logarithm of one plus the duration (in months) of the loan. Interest Rate (%) is the interest earned on a loan. Loan Amount is the logarithm of one plus loan amount. Risk-Adjusted Interest Rate is the risk-adjusted interest rate of a loan. Marital Status is a dummy for borrowers where 1 if Married, and 0 otherwise. High Education is a dummy for borrowers where 1 is Masters or Above, and 0 otherwise. Male is a dummy for borrowers where 1 if Male, and 0 otherwise. Self Employed is a dummy variable for borrowers where 1 if self-employed, and 0 otherwise. Borrower Risk is the logarithm of one plus borrower risk score that varies from 1 to 245.

	(1)	(2)	(3)	(4)	(5)
Variable	Mean	Std	P25	P50	P75
Panel A: Investor-Bid Level (Obs = 23,687,997)					
Auto Bidding	0.89	0.31	1.00	1.00	1.00
Share Auto Bidding	0.82	0.27	0.77	0.95	0.99
Share Auto Bidding^2	0.74	0.32	0.59	0.89	0.98
General Switch	0.04	0.20	0.00	0.00	0.00
Switch to Auto	0.02	0.15	0.00	0.00	0.00
Switch to Manual	0.02	0.14	0.00	0.00	0.00
Sharpe Ratio	0.16	0.03	0.15	0.16	0.17
Profile Age	4.67	1.41	3.91	4.99	5.72
Bid Volume	5.68	1.48	4.78	5.83	6.73
Decision Time	0.07	0.39	0.00	0.00	0.00
Relative Decision Time	0.49	0.29	0.24	0.49	0.74
Panel B: Investor Level (Obs = 68,020)					
Auto Bidding	0.82	0.25	0.77	0.94	0.98
Share Auto Bidding	0.73	0.29	0.58	0.85	0.96
Share Auto Bidding^2	0.64	0.32	0.40	0.75	0.92
General Switch	0.06	0.08	0.01	0.03	0.08
Switch to Auto	0.04	0.05	0.01	0.02	0.05
Switch to Manual	0.02	0.04	0.00	0.01	0.03
Sharpe Ratio	0.16	0.03	0.15	0.16	0.17
Profile Age	3.25	1.25	2.26	3.31	4.24
Bid Volume	4.03	1.34	3.05	4.03	4.99
Decision Time	0.13	0.24	0.02	0.04	0.13
Relative Decision Time	0.50	0.06	0.47	0.50	0.53
Panel C: Loan Level, Obs = 841,903					
Maturity	3.40	0.43	3.22	3.61	3.61
Interest Rate	0.11	0.01	0.10	0.10	0.11
Loan Amount	10.98	0.83	10.45	11.16	11.58
Risk-Adjusted Interest Rate	0.16	0.12	0.15	0.15	0.15
Panel D: Borrower Level (Obs = 803,571)					

Married	0.58	0.49	0.00	1.00	1.00
High Education	0.01	0.12	0.00	0.00	0.00
Male	0.66	0.47	0.00	1.00	1.00
Self Employed	0.18	0.38	0.00	0.00	0.00
Borrower Risk	4.23	0.18	4.20	4.20	4.20

Table 2: Descriptive statistics: manual users vs automatic users

This table represents the t-test for the mean difference between manual users and automatic users. The variable definitions are in Table 1. *** indicate statistical significant at the 1% level.

	Mean	Std	Mean	Std	Difference
	Only Manual Bids		**Only Automatic Bids**		**Manual vs Auto**
Sharpe Ratio	0.15	0.06	0.16	0.02	-0.01***
Profile Age	3.81	1.57	4.78	1.35	-0.97***
Bid Volume	4.73	1.64	5.80	1.41	-1.06***
Decision Time	0.46	0.93	0.03	0.21	0.43***
Relative Decision Time	0.52	0.29	0.49	0.29	0.03***
Maturity	3.13	0.59	3.54	0.24	-0.40***
Interest Rate	0.12	0.02	0.10	0.01	0.01***
Loan Amount	10.99	0.89	11.44	0.58	-0.45***
Loan Performance	0.16	0.19	0.16	0.02	0.00***
Borrower Risk	4.37	0.39	4.21	0.03	-0.05***
Married	0.70	0.46	0.67	0.47	0.03***
High Education	0.02	0.14	0.02	0.15	-0.00***
Male	0.73	0.44	0.64	0.48	0.09***
Self Employed	0.28	0.45	0.20	0.40	0.09***

Table 3: Determinants of Automatic Biddings

The dependent variable in all columns is a binary variable that equals 1 if the investor bids using the auto bidding tool, and 0 if manual bid. Independent variables are defined in Table 1. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. ***, and ** indicate statistical significant at the 1% and 5 % level, respectively.

	(1) Auto Bidding	(2) Auto Bidding	(3) Auto Bidding	(4) Auto Bidding
Sharpe Ratio_{t-1}	-0.160***	-0.035***	-0.013***	-0.002**
	(0.006)	(0.001)	(0.001)	(0.001)
Share Auto Bidding_{t-1}	1.542***	0.406***	0.469***	0.388***
	(0.001)	(0.001)	(0.001)	(0.001)
Share Auto Bidding^2_{t-1}	-0.717***	-0.227***	-0.311***	-0.268***
	(0.001)	(0.001)	(0.001)	(0.001)
Maturity_t	0.021***	0.006***	0.009***	
	(0.000)	(0.001)	(0.001)	
Interest Rate_t	0.094***	-1.376***	-1.511***	
	(0.006)	(0.018)	(0.018)	
Loan Amount_t	0.001***	0.017***	0.017***	
	(0.000)	(0.000)	(0.000)	
Borrower Risk_t	-0.248***			
	(0.001)			
Married_t	-0.004***			
	(0.000)			
High Education_t	0.000			
	(0.000)			
Male_t	-0.000***			
	(0.000)			
Self Employed_t	0.003***			
	(0.000)			
Investor Fixed Effect	No	No	Yes	Yes
Borrower Fixed Effect	No	Yes	Yes	No
Hour of Bid Fixed Effect	Yes	Yes	Yes	Yes
Loan Fixed Effect	No	No	No	Yes
Observations	72,791,534	72,773,059	72,747,717	72,744,977
Adjusted R2	0.605	0.917	0.925	0.945

Table 4: Switching mode of bidding

The dependent variables are General Switching, Switch to Automatic, Switch to Manual in columns (1), (2) and (3), respectively. These variables are defined as switching indicators for investors. General Switch is a binary variable that equals one if the investor attempted a general switch (manual or automatic), and 0 otherwise. Switch to Automatic is a binary variable that equals 1 if the investor switch to an automated mode, and 0 otherwise. Switch to Manual is a binary variable that equals 1 if the investor switch to a manual mode, and 0 otherwise. Independent variables are defined in Table 1. All regressions include a constant term. Robust standard errors are in parentheses and clustered at investor within a loan level. ***, and ** indicate statistical significant at the 1% and 5% level, respectively.

	(1) General Switching	(2) Switch to Automatic	(3) Switch to Manual
Sharpe Ratio$_{t-1}$	0.004**	-0.014***	0.018***
	(0.002)	(0.002)	(0.002)
Share Auto Bidding$_{t-1}$	0.350***	-0.389***	0.739***
	(0.007)	(0.006)	(0.004)
Share Auto Bidding$^2_{t-1}$	-0.361***	0.247***	-0.608***
	(0.007)	(0.006)	(0.004)
Sharpe Ratio$_{t-1}$* Share Auto Bidding$_{t-1}$	0.486***	1.204***	-0.717***
	(0.042)	(0.037)	(0.026)
Sharpe Ratio$_{t-1}$ * Share Auto Bidding$^2_{t-1}$	-0.464***	-1.591***	1.126***
	(0.044)	(0.038)	(0.027)
Investor Fixed Effect	Yes	Yes	Yes
Hour of Bid Fixed Effect	Yes	Yes	Yes
Loan Fixed Effect	Yes	Yes	Yes
Observations	23,465,087	23,465,087	23,465,087
Adjusted R^2	0.169	0.110	0.306

Table 5: Inattention and time pressure before changing mode of investing

The dependent variables are General Switching, Switch to Automatic, Switch to Manual in columns 1-3, respectively. These variables are defined as switching indicators for investors. General Switch is a binary variable that equals 1 if the investor attempted a general switch (manual or automatic), and 0 otherwise. Switch to Automatic is a binary variable that equals 1 if the investor switch to an automated mode, and 0 otherwise. Switch to Manual is a binary variable that equals 1 if the investor switch to a manual mode, and 0 otherwise. Independent variables are defined in Table 1. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. ***, and * indicate statistical significant at the 1% and 10 % level, respectively.

	(1) Switch to Automatic	(2) Switch to Manual	(3) Switch to Automatic	(4) Switch to Manual
Sharpe Ratio_{t-1}	-0.125***	0.071***	-0.136***	0.056***
	(0.014)	(0.008)	(0.016)	(0.008)
Share Auto Bidding_{t-1}	-0.380***	0.727***	-0.380***	0.727***
	(0.008)	(0.005)	(0.008)	(0.005)
Share Auto Bidding_{t-1}^2	0.237***	-0.598***	0.237***	-0.598***
	(0.007)	(0.005)	(0.007)	(0.005)
Sharpe Ratio_{t-1}* Share Auto Bidding_{t-1}	1.310***	-0.746***	1.310***	-0.747***
	(0.050)	(0.031)	(0.050)	(0.031)
Sharpe Ratio_{t-1} * Share Auto Bidding_{t-1}^2	-1.597***	1.106***	-1.597***	1.107***
	(0.045)	(0.029)	(0.045)	(0.029)
Decision Time_{t-1}	0.069***	-0.045***	0.069***	-0.045***
	(0.001)	(0.000)	(0.001)	(0.000)
Sharpe Ratio_{t-1}* Decision Time_{t-1}	0.053***	-0.025***	0.052***	-0.026***
	(0.006)	(0.003)	(0.006)	(0.003)
Relative Decision Time_{t-1}			-0.002	-0.001
			(0.002)	(0.001)
Sharpe Ratio_{t-1}* Relative Decision Time_{t-1}			0.020*	0.029***
			(0.012)	(0.007)
Investor Fixed Effect	Yes	Yes	Yes	Yes
Hour of Bid Fixed Effect	Yes	Yes	Yes	Yes
Loan Fixed Effect	Yes	Yes	Yes	Yes
Observations	23,465,087	23,465,087	23,465,087	23,465,087
R^2	0.168	0.342	0.168	0.342

Table 6: Experience before changing mode of investing

The dependent variables are General Switching, Switch to Automatic, Switch to Manual in columns 1-3, respectively. These variables are defined as switching indicators for investors. General Switch is a binary variable that equals 1 if the investor attempted a general switch (manual or automatic), and 0 otherwise. Switch to Automatic is a binary variable that equals 1 if the investor switch to an automated mode, and 0 otherwise. Switch to Manual is a binary variable that equals 1 if the investor switch to a manual mode, and 0 otherwise. Independent variables are defined in Table 1. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. ***, and * indicate statistical significant at the 1% and 10% level, respectively.

	(1) Switch to Automatic	(2) Switch to Manual	(3) Switch to Automatic	(4) Switch to Manual
Sharpe Ratio$_{t-1}$	-0.132***	0.083***	-0.156***	0.072***
	(0.016)	(0.010)	(0.019)	(0.011)
Share Auto Bidding$_{t-1}$	-0.376***	0.750***	-0.355***	0.752***
	(0.008)	(0.005)	(0.008)	(0.005)
Share Auto Bidding$^2_{t-1}$	0.235***	-0.634***	0.207***	-0.640***
	(0.007)	(0.005)	(0.007)	(0.005)
Sharpe Ratio$_{t-1}$* Share Auto Bidding$_{t-1}$	1.289***	-0.892***	1.146***	-0.910***
	(0.050)	(0.032)	(0.049)	(0.032)
Sharpe Ratio$_{t-1}$ * Share Auto Bidding$^2_{t-1}$	-1.584***	1.318***	-1.432***	1.336***
	(0.045)	(0.030)	(0.044)	(0.030)
Decision Time$_{t-1}$	0.069***	-0.045***	0.069***	-0.045***
	(0.001)	(0.000)	(0.001)	(0.000)
Sharpe Ratio$_{t-1}$* Decision Time$_{t-1}$	0.053***	-0.027***	0.053***	-0.026***
	(0.006)	(0.003)	(0.006)	(0.003)
Profile Age$_{t-1}$	-0.001	0.006***		
	(0.001)	(0.000)		
Sharpe Ratio$_{t-1}$* Profile Age$_{t-1}$	0.006	-0.005*		
	(0.004)	(0.002)		
Bid Volume$_{t-1}$			0.000	0.005***
			(0.001)	(0.000)
Sharpe Ratio$_{t-1}$* Bid Volume$_{t-1}$			0.019***	0.004
			(0.004)	(0.003)
Investor Fixed Effect	Yes	Yes	Yes	Yes
Hour of Bid Fixed Effect	Yes	Yes	Yes	Yes
Loan Fixed Effect	Yes	Yes	Yes	Yes
Observations	23,465,08 7	23,465,08 7	23,465,08 7	23,465,087
R^2	0.168	0.342	0.168	0.342

Appendix

Table A1: Decision to Switch (Restricted Bid Attempts)

The dependent variable in column 1 is General Switch and is a binary variable that equals 1 if the investor attempted a general switch (manual or automatic), 0 otherwise. The dependent variable in column 2 is Switch to Automatic and is a binary variable that equals 1 if the investor switch to an automated mode, and 0 otherwise. In column 3, the dependent variable is Switch to Manual and is a binary variable that equals 1 if the investor switch to a manual mode, and 0 otherwise. Independent variables are defined in Table 1. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. ***, and ** indicate statistical significant at the 1% and 5% level, respectively.

Panel A: Bid Attempts Restricted to no less than 10 Bids			
	(1) General Switching	(2) Switch to Automatic	(3) Switch to Manual
Sharpe Ratio$_{t-1}$	0.003**	-0.015***	0.018***
	(0.002)	(0.002)	(0.002)
Share Auto Bidding$_{t-1}$	0.350***	-0.390***	0.740***
	(0.007)	(0.005)	(0.004)
Share Auto Bidding$^2_{t-1}$	-0.361***	0.250***	-0.611***
	(0.007)	(0.005)	(0.004)
Sharpe Ratio$_{t-1}$* Share Auto Bidding$_{t-1}$	0.504***	1.232***	-0.728***
	(0.039)	(0.029)	(0.026)
Sharpe Ratio$_{t-1}$ * Share Auto Bidding$^2_{t-1}$	-0.481***	-1.619***	1.138***
	(0.041)	(0.031)	(0.027)
Investor Fixed Effect	Yes	Yes	Yes
Hour of Bid Fixed Effect	Yes	Yes	Yes
Loan Fixed Effect	Yes	Yes	Yes
Observations	23,456,556	23,456,556	23,456,556
Adjusted R^2	0.168	0.110	0.306

Table A2: Decision to Switch (Restricted Bid Attempts)

The dependent variable in column 1 is General Switch and is a binary variable that equals 1 if the investor attempted a general switch (manual or automatic), and 0 otherwise. The dependent variable in column 2 is Switch to Automatic and is a binary variable that equals 1 if the investor switch to an automated mode, and 0 otherwise. In column 3 the dependent variable is Switch to Manual and is a binary variable that equals 1 if the investor switch to a manual mode, and 0 otherwise. Sharpe Ratio is the risk-adjusted returns weighted by the bid amount. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. *** indicate statistical significant at the 1% level.

Panel B: Bid Attempts Restricted to no less than 20 Bids			
	(1) General Switching	(2) Switch to Automatic	(3) Switch to Manual
Sharpe Ratio$_{t-1}$	0.004***	-0.014***	0.019***
	(0.002)	(0.002)	(0.002)
Share Auto Bidding$_{t-1}$	0.368***	-0.375***	0.743***
	(0.007)	(0.005)	(0.004)
Share Auto Bidding$^2_{t-1}$	-0.375***	0.238***	-0.613***
	(0.007)	(0.005)	(0.004)
Sharpe Ratio$_{t-1}$* Share Auto Bidding$_{t-1}$	0.461***	1.199***	-0.738***
	(0.039)	(0.029)	(0.026)
Sharpe Ratio$_{t-1}$ * Share Auto Bidding$^2_{t-1}$	-0.443***	-1.587***	1.145***
	(0.041)	(0.030)	(0.027)
Investor Fixed Effect	Yes	Yes	Yes
Hour of Bid Fixed Effect	Yes	Yes	Yes
Loan Fixed Effect	Yes	Yes	Yes
Observations	23,398,504	23,398,504	23,398,504
Adjusted R^2	0.169	0.109	0.306

Table A3: Switching mode of bidding

This table presents results using alternative performance measurement. The dependent variables are General Switching, Switch to Automatic, Switch to Manual in columns 1, 2 and 3, respectively. These variables are defined as switching indicators for investors. General Switch is a binary variable that equals one if the investor attempted a general switch (manual or automatic), 0 otherwise. Switch to Automatic is a binary variable that equals 1 if the investor switch to an automated mode, and 0 otherwise. Switch to Manual is a binary variable that equals 1 if the investor switch to a manual mode, and 0 otherwise. Unique Investments is the logarithm of one plus the number of unique successful investments. Share Auto Bidding is the cumulative percentage of using automated bidding by investors. Share Auto Bidding^2 is the quadratic form of the share of automatic bidding. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. *** indicate statistical significant at the 1% level.

	(1) General Switching	(2) Switch to Automatic	(3) Switch to Manual
Unique Investments$_{t-1}$	-0.003***	-0.013***	0.009***
	(0.000)	(0.000)	(0.000)
Share Auto Bidding$_{t-1}$	0.165***	-0.435***	0.600***
	(0.005)	(0.003)	(0.002)
Share Auto Bidding$^2_{t-1}$	-0.246***	0.134***	-0.379***
	(0.004)	(0.003)	(0.002)
Unique Investments$_{t-1}$* Share Auto Bidding$_{t-1}$	0.048***	0.054***	-0.005***
	(0.001)	(0.001)	(0.000)
Unique Investments$_{t-1}$ * Share Auto Bidding$^2_{t-1}$	-0.041***	-0.037***	-0.004***
	(0.001)	(0.001)	(0.000)
Investor Fixed Effect	Yes	Yes	Yes
Hour of Bid Fixed Effect	Yes	Yes	Yes
Loan Fixed Effect	Yes	Yes	Yes
Observations	23,532,002	23,532,002	23,532,002
Adjusted R^2	0.166	0.119	0.304

Table A4: Loan Level Switching

This table examines the relationship between loan performance and three switching indicators. General, Automatic and Manual Switchers are the number of general, automatic, manual switchers in a loan. Loan Performance is the risk-adjusted returns of a loan. Loan Share Auto Bidding is the percentage of using automated bidding in a loan. Loan Maturity is the logarithm of one plus the duration (in months) of the loan. Credit Risky is a dummy where 1 is HR or E loan risk, and 0 otherwise. Loan Amount is the logarithm of one plus loan amount. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. ***, and ** indicate statistical significant at the 1% and 5% level, respectively.

	(1) General Switchers	(2) Automatic Switchers	(3) Manual Switchers
$\text{Risk} - \text{Adjusted Interest Rate}_j$	-0.044	0.121***	-0.180**
	(0.043)	(0.047)	(0.089)
$\text{Loan Share Auto Bidding}_j$	-0.537***	0.942***	-1.504***
	(0.022)	(0.022)	(0.017)
Loan Maturity_j	-0.258***	-0.608***	0.077***
	(0.014)	(0.015)	(0.013)
Credit Risky_j	0.262***	0.481***	0.011
	(0.033)	(0.038)	(0.034)
Loan Amount_j	0.646***	0.448***	0.531***
	(0.009)	(0.010)	(0.009)
Hour of Listing Fixed Effect	Yes	Yes	Yes
Observations	17,784	17,784	17,784
Adjusted R^2	0.235	0.249	0.378

Table A5: Loan Level Automatic Shares

This table examines the relationship between loan share auto bidding and loan performance indicators. Risk is the log of one plus the risk score (1 – 245) of a loan. Return is the interest rate of a loan. Loan Performance is the risk-adjusted return of a loan. Loan Share Auto Bidding is the percentage of using automated bidding in a loan. Loan Maturity is the logarithm of one plus the duration (in months) of the loan. Credit Risky is a dummy where 1 is HR or E loan risk, and 0 otherwise. Loan Amount is the logarithm of one plus loan amount. Percent Funded is the percentage funded of a loan. All regressions include a constant term. Robust standard errors are in parentheses and clustered investor within a loan level. *** indicate statistical significant at the 1% level.

	(1) Loan Risk	(2) Interest Rate	(3) Risk-Adjusted Interest Rate
Loan Share Auto Bidding$_j$	-0.018***	-0.011***	-0.011***
	(0.002)	(0.000)	(0.001)
Loan Maturity$_j$	-0.065***	0.008***	0.003
	(0.006)	(0.000)	(0.010)
Credit Risky$_j$	0.888***	0.018***	-0.108***
	(0.011)	(0.001)	(0.010)
Loan Amount$_j$	0.029***	0.001***	-0.009***
	(0.003)	(0.000)	(0.003)
Percent Funded$_j$	-0.156	-0.014***	0.011
	(0.119)	(0.005)	(0.018)
Hour of Listing Fixed Effect	Yes	Yes	Yes
Observations	17,784	17,784	17,784
Adjusted R^2	0.582	0.206	0.024